

Research Review - Reconfigurable Intelligent Surfaces

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Research question

Reconfigurable intelligent surfaces (RIS) are a novel technology in data communications, with many papers on the topic appearing in the last two years. The topic is interesting to all three of us, with our different Electrical Engineering specializations. RIS touches on both the digital and the analog domain, as well as electromagnetism. Therefore we will write a literature review on the following research question:

How do different reconfigurable intelligent surface configurations affect communication link strength?

The following sub questions must be answered to answer the main research question above:

- What is a Reconfigurable Intelligent Surface (RIS)?
- What are the different RIS configurations?
- How do the physics behind different RIS configurations work?
- How can the different RIS configurations be applied in a communication system context?
- What are the benefits and disadvantages of applying RIS in a communication system?

Structure

Since the literature review will incorporate various articles from research groups working on RIS, the Methods and Results sections will contain a comparison of various methods and results. The following structure will be used:

1. Abstract
2. Introduction
3. Comparison of methods
4. Comparison of results
5. Conclusion
6. Discussion
7. Bibliography

Statement on the use of generative AI

When writing this research proposal, generative AI was used for brainstorming on topics to write about, and it was used to find more literature. AI was not used to dictate or correct the written text.

1 Introduction

The rapid evolution of wireless data communication is approaching a transition toward sixth-generation (6G) networks, which demand ultra-high data rates, low latency, and improved reliability. To meet these requirements, future systems have to rely on high-frequency bands such as the millimeter-wave (mmWave) spectrum. While these frequencies enable much higher bandwidths, they also introduce physical challenges, including severe path loss and high susceptibility to attenuation by obstacles. As a result, conventional communication systems, which treat the propagation environment as an uncontrollable medium, face performance limitations. Reconfigurable intelligent surfaces (RIS) have emerged as a promising solution to these challenges by enabling dynamic control over electromagnetic wave propagation.

An RIS consists of a large array of engineered electromagnetic elements whose reflection properties can be electronically adjusted in real time. By manipulating the phase, amplitude, or polarization of incident waves, RIS can redirect signals, enhance constructive interference, and improve signal-to-noise ratio (SNR) at the receiver. This capability transforms the wireless channel from a passive transmission

medium into a programmable, controllable component of the communication system. Because RIS technology combines principles from electromagnetics, analog circuit design, and digital signal processing, it represents a multidisciplinary innovation within electrical engineering.

Current research primarily distinguishes between two architectural approaches: passive and active RIS configurations. Passive RIS, which dominates early theoretical studies, relies on nearly lossless reflecting elements that do not require integrated power amplifiers, providing high energy efficiency and hardware simplicity. Passive designs on the other hand, are inherently constrained by the double-fading effect, in which signal attenuation occurs along both the transmitter-to-surface and surface-to-receiver paths. In contrast, active RIS incorporates amplification components that can boost reflected signals, thereby mitigating double-fading and extending the communication range. While active configurations can significantly enhance SNR, they introduce trade-offs related to increased power consumption, additional noise, and hardware complexity.

Beyond the passive-active distinction, recent studies explore hybrid and multifunctional RIS frameworks. Surfaces capable of switching between passive reflection, active amplification, and off states provide flexibility in dynamic communication environments. Furthermore, unified frameworks combining RIS enhanced communication with ambient radio-frequency (RF) energy harvesting have been proposed to improve system sustainability and support low-power Internet of Things (IoT) devices. In all configurations, advanced control strategies, particularly adaptive beamforming and real-time phase optimization, play an important role in maintaining link robustness in obstructed or rapidly changing environments for high-frequency communication systems.

Experimental and simulation-based studies have investigated RIS performance in industrial indoor environments and mmWave systems, demonstrating that positioning, phase tuning, and configuration type significantly influence received power and overall link strength. Despite the growing body of literature, comparative analyses synthesizing how architectural differences fundamentally affect communication link strength remain limited.

This literature review therefore addresses the research question: How do different reconfigurable intelligent surface configurations affect communication link strength? By comparing underlying physical principles, implementation strategies, and reported performance metrics, this review aims to clarify the benefits and limitations of passive, active, and hybrid RIS configurations and assess their suitability for future high-frequency wireless communication systems.

2 Comparison of Methods

To study the effectivity of a RIS configuration, two important metrics are the Bit Error Rate (BER), and the Signal to Noise Ratio (SNR). To obtain these and other metrics for a certain configuration, simulation and experimental setups can be used. All papers included in this literature review used simulation as a means to quantify performance of RIS configurations.

Radpour et al. simulated an RIS-panel with 127 square RIS elements with 6.6mm sides [1]. An absorber was placed between a transmitting horn antenna and the omnidirectional receiving antenna. The total path from the transmit antenna to the RIS panel and then to the receiver was always less than 2m. This setup is similar to the setup used by Amri et al. in [2], since both simulated an indoor scenario with low transmission distances. Amri et al. used a bigger RIS-panel, with 256 RIS elements, but with a lower cell dimension. Instead of RIS elements with 6.6mm sides, Amri et al. used elements with sides of 36.2mm. These larger elements were paired with a lower frequency of 5.8 GHz, a frequency from the free medical and industrial ISM band. Radpour et al. opted to use a higher frequency of 23.8 GHz, from the mmWave band. This makes their results relevant for 5G and 6G applications of RIS surfaces.

These two setups have two limitations. For one, they only provide data on short range indoor scenarios. Secondly, they both use the assumption that the direct link is completely blocked, which inflates the relative RIS performance. This is in contrast with the setup simulated by Enahoro et al. in [3]. In this setup, the direct link is present as well. Enahoro et al. simulated the RIS link for multiple distances, from 100m to 1000m, and varied the number of receiving antennas from 10 to 1000, and varied the number of RIS elements from 100 to 1000, although only results for between 100 and 200 RIS elements are published.

	without RIS	Passive	Active
sum-rate	2.98 bps/Hz	13.80 bps/Hz	33.39 bps/Hz
sum-rate gain	0%	363%	1020%

Table 1: Results of sum-rate analysis in [4]

	Passive	Active
number of elements	$N = 256$	$N = 256$
base station to RIS path loss	$\varrho_g^2 = -70 \text{ dB}$	$\varrho_g^2 = -70 \text{ dB}$
RIS to user path loss	$\varrho_f^2 = -70 \text{ dB}$	$\varrho_f^2 = -70 \text{ dB}$
RIS noise power	$\sigma_v = 0 \text{ dBm}$	$\sigma_v = -100 \text{ dBm}$
static noise power	$\sigma = -100 \text{ dBm}$	$\sigma = -100 \text{ dBm}$
Transmit power	$P_{tx} = 2 \text{ W}$	$P_{tx} = 1 \text{ W}$
amplification power	$P_A = 0 \text{ W}$	$P_A = 1 \text{ W}$
total power consumption	$P_{tot} = 2 \text{ W}$	$P_{tot} = 2 \text{ W}$
SNR	$\gamma_{passive} \approx 39.0 \text{ dB}$	$\gamma_{active} \approx 79.0 \text{ dB}$

Table 2: Parameters used in [4]
to asymptotically evaluate SNR performance.

3 Comparison of Results

In this section, we compare the reported performance of different RIS configurations with respect to communication link strength. We focus on key metrics such as Signal-to-Noise-Ratio (SNR), Bit Error Rate (BER), received power, and achievable sum rate across various studies.

3.1 Results from Active RIS

Zhang et al. [4] introduced the concept of active RIS in order to overcome the multiplicative fading effect. The core idea of active RIS is that the RIS elements, in addition to providing control over the phase of the reflected signal also actively amplify the signal. In order to estimate the capacity gain of active RIS, they have developed an analytical expression for the achievable SNR in a single-user single-input single-output case (SU-SISO). They used practical values for parameters like noise power, transmit power and number of RIS elements and showed that for the same total power consumption, active RIS outperforms passive RIS by a factor of 10,000 in terms of SNR. The parameters and results are summarized in table 2. They also generalized the analysis to a more applicable multiple-user, multiple-input, single-output case (MU-MISO). Here, an important metric is the sum-rate. The sum rate follows from the well known Shannon's capacity theorem and it shows how spectrally efficient the system could theoretically be.

$$R_{sum} = \log_2(1 + SNR) \text{ bps/Hz} \quad (1)$$

The sum rate was analyzed for two different scenarios, one with a strong direct link and one without a strong direct link. The x-distance to the base station was set to 300 meters, the transmit power at 10 dBm and four users were serviced. It was found that in the first scenario, active RIS could achieve more than a 1000% increase in the sum rate. The results are summarized in table 1.

Another important result from their work is the effect of increasing the number of elements on the sum rate. It was found that active RIS benefits a lot more from increasing the amount of elements than passive RIS. Going from 100 to 900 elements, active RIS had an increase of 15.32 bps/Hz compared to only 7.21 bps/Hz for the passive RIS. An interesting observation can be made when a strong direct link was added, active RIS practically had the same increase in sum rate, while the increase in sum rate for passive RIS was only half.

3.2 Benefits of Adaptive Beamforming

The effectiveness of a passive RIS is not just dependent on its hardware but also on the control algorithms used for beamforming. Amri et al.[2] validated a 1-bit passive RIS using a compressive sensing-based adaptive beamforming algorithm. Their results indicate that even a simple 1-bit phase resolution can

drastically improve link quality indoors. For QPSK modulation, the BER was reduced from 4.69×10^{-6} to 1.44×10^{-8} when beamforming was active.

A significant advantage of this algorithm is the reduction in training overhead. They found that reducing training patterns from 64 to 16 resulted in an SNR loss of only about 1 dB, which suggest that passive RIS can maintain high link strength even in dynamic environments with low signaling overhead. Similarly, Radpour et al.[1] analyzed the beam footprint, finding that focus areas are much narrower in azimuth than radially. This implies that that beamforming must be highly precise to keep the signal locked onto mobile users. Without these beamforming strategies, the link strength would drop significantly as the user moves.

3.3 RIS Operation at mmWave Frequencies

Operating in the millimeter-wave (mmWave) band (20-30 GHz) presents a unique challenges due to severe path loss and blockages. Radpour et al.[1, 5] and Enahoro et al.[3] specifically target these high frequencies. In an industrial NLoS scenario where the direct link is blocked by absorbers, Radpour et al.[1] showed that an active RIS can maintain an SNR above 40 dB. While passive elements struggle with the attenuation at these frequencies, switching to an active mode with a 1.25 amplification factor resulted in a power improvement of at least 4 dB over the passive reflective mode. The ability to bypass blockages with such high signal fidelity is a major step for industrial application where machinery often obstructs the direct line of sight.

Enahoro et al.[3] expanded on this by evaluating a unified framework at 28 GHz. Their simulations shows that active RIS enhanced systems could increase throughput from 2.6 Gbps to 6.8 Gbps. Furthermore, they reported a significant reduction in latency, dropping from 80 ms to 35 ms. These mmWave results are particularly impressive because they also demonstrate an improvement in overall energy efficiency, despite the power consumptions of active elements. The overall system energy efficiency rose to 0.82 compared to 0.44 for a standard direct link.

3.4 Summary of Link Performance Metrics

This table provides a clear overview of the key results discussed in this review, allowing for a direct comparison of how different configurations impact communication link strength.

Table 3: Comparison of Reported RIS Performance Metrics

Reference	RIS Type	Frequency	Main Metric	Reported Result/Gain
Amri et al. [2]	Passive (256 el.)	5.8 GHz	BER / SNR	12.65 dB SNR gain; BER reduced to 10^{-8}
Radpour et al. [1]	Active (127 el.)	23.8 GHz	Received Power	≥ 4 dB gain over reflective mode
Zhang et al. [4]	Active vs Passive	N/A (Sim)	Sum Rate	130% gain (Active) vs 22% (Passive)
Enahoro et al. [3]	Active (100-1000 el.)	28 GHz	Throughput	6.8 Gbps (Active) vs 4.6 Gbps (Passive)

In conclusion, the evidence from these studies suggest that while passive RIS is a highly energy-efficient and cost-effective solution for short-range indoor scenarios, it is much weaker then active options when it comes to mmWave and multi-user environments. Passive RIS relies heavily on large element counts and smart beamforming to overcome multiplicative fading. By contrast, active surfaces introduce amplification to solve this issue directly. Both architectures play a critical role, but their deployment will depend strictly on the specific link requirements and power constraints of future wireless networks. Ultimately, creating a robust 6G infrastructure will likely require a heterogeneous approach, seamlessly integrating both passive and active configurations to optimize the balance between maximum coverage area, acceptable latency, and total power consumption.

4 Conclusion and Discussion

References

- [1] Hamed Radpour et al. “Reconfigurable intelligent surface for industrial automation: mmWave propagation measurement, simulation, and control algorithm requirements”. In: *2024 IEEE 35th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*. IEEE, Sept. 2024, pp. 1–7.
- [2] Muhammad Miftahul Amri, Nguyen Minh Tran, and Kae Won Choi. “Reconfigurable intelligent surface-aided wireless communications: Adaptive beamforming and experimental validations”. In: *IEEE Access* 9 (2021), pp. 147442–147457.
- [3] Sunday Enahoro et al. “Unified framework for RIS-enhanced wireless communication and ambient RF energy harvesting: Performance and sustainability analysis”. en. In: *Technologies (Basel)* 13.6 (June 2025), p. 244.
- [4] Zijian Zhang et al. “Active RIS vs. Passive RIS: Which Will Prevail in 6G?” In: *IEEE Trans. Commun.* 71.3 (Mar. 2023), pp. 1707–1725.
- [5] Hamed Radpour et al. “Active reconfigurable intelligent surface for the millimeter-wave frequency band: Design and measurement results”. In: *2024 IEEE Wireless Communications and Networking Conference (WCNC)*. IEEE, Apr. 2024.